

Data-Quality & EDA Insight Memo

Project FORESIGHT · NorthBay Living · Zidio Development

Data provenance note: The project brief says four simulated client extracts should be provided. Those raw files were not available in the supplied dashboard materials. This prototype therefore uses a schema-compatible synthetic dataset generated with a fixed seed. Findings below describe the prototype dataset.

1. Data Quality — What We Found and How It Was Handled

Extract	Issue	Handling
sku_master	203 input rows; 3 duplicate SKU rows; inconsistent category labels	Removed duplicate SKU rows; normalised category labels.
sku_master	4 missing unit_cost values	Imputed with category median.
calendar	731 daily rows; empty promo_event values	No structural issue; empty promo means no promotion.
sales_daily	120 exact duplicates; 15 negative units_sold	Removed exact duplicates; clipped negative units to 0 and flagged for review.
sales_daily	250 missing units_sold	Filled with 0 and retained a missingness flag.
inventory_snapshots	10 missing lead_time_days	Imputed with each SKU's median lead time.

2. Business-Relevant Insights

- Insight 1 — Promotions are a material demand signal.** Average weekly unit demand is 152 in non-promo weeks versus 263 in promo weeks, a lift of about 73%. Inventory buffers should be planned before known promotions.
- Insight 2 — Demand is concentrated in top movers.** A small group of SKUs accounts for a disproportionate share of unit demand. These products deserve tighter stockout monitoring because their commercial impact is larger.
- Insight 3 — Calendar effects matter.** The synthetic data contains meaningful seasonal, holiday and promotional patterns, supporting calendar-aware features instead of relying on a single flat demand curve.
- Insight 4 — A declining tail exists.** Several SKUs show a large recent decline versus the prior eight weeks, making them candidates for markdown review even if their absolute sales volume is not large.
- Insight 5 — Risk should be prioritised by money, not SKU count alone.** The risk layer ranks SKUs by rupee value at stake so operations can focus on the few decisions with the largest financial exposure.

3. Modelling Implication

The forecasting workflow uses weekly SKU demand, leakage-safe lags/rolling features, calendar signals, a seasonal-naive benchmark and rolling-origin validation. The current run has four folds and the LightGBM model averages WAPE 0.199 versus 0.236 for the seasonal-naive baseline.